

Arithmetic Interference and Geometric Rigidity: A Structural Approach to the Riemann Hypothesis

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Abstract

We present a structural reduction of the Riemann Hypothesis based on the decomposition of the Riemann zeta function into three independent arithmetic components $\zeta_A, \zeta_B, \zeta_C$, induced by a tetrahedral partition of the primes. We prove that a non-trivial zero off the critical line would require a simultaneous balance of amplitudes and derivatives (Wronskian condition) between these components. By establishing a *Quantitative Drift Theorem* (Lemma 2'), we show that outside the critical line $\Re(s) = 1/2$, the amplitudes of the components diverge asymptotically due to the disjointness of their prime supports. Furthermore, we demonstrate via *Wronskian Rigidity* (Lemma 3) that the logarithmic derivatives cannot synchronize. Consequently, the destructive interference required for a zero is analytically impossible for $\Re(s) \neq 1/2$.

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1 Introduction

The Riemann Hypothesis (RH) asserts that all non-trivial zeros of the Riemann zeta function $\zeta(s)$ lie on the critical line $\Re(s) = 1/2$. While classical approaches focus on global analytic estimates, this paper proposes a *geometric-dynamic* perspective. We model the zeta function as a superposition of three independent arithmetic waves, motivated by a quaternion-induced partition of the primes $\mathbb{P}$.

We define the global interference function:

$$\Phi(s) := \alpha\zeta_A(s) + \beta\zeta_B(s) + \gamma\zeta_C(s), \quad (1)$$

where ζ_X are partial Euler products over disjoint prime sets. The central argument is that a zero of $\Phi(s)$ requires a precise geometric closure (a triangle in $\mathbb{C}$) which is structurally unstable under analytic continuation away from the axis of symmetry.

2 Arithmetic Decomposition

Let $\mathbb{P} = \{2, 3, 5, \dots\}$ be the set of prime numbers. We assume a disjoint partition

$$\mathbb{P} = \mathbb{P}_A \dot{\cup} \mathbb{P}_B \dot{\cup} \mathbb{P}_C, \quad (2)$$

such that each set has positive natural density. For $\Re(s) > 1$, we define the components:

$$\zeta_X(s) := \prod_{p \in \mathbb{P}_X} (1 - p^{-s})^{-1}, \quad X \in \{A, B, C\}. \quad (3)$$

By analytic continuation, these functions are defined on $\mathbb{C} \setminus \{1\}$.

3 The Global Obstruction: Amplitude Drift

A necessary condition for $\Phi(s_0) = 0$ is the triangle inequality for the magnitudes: $|\zeta_A(s_0)| \approx |\zeta_B(s_0)| \approx |\zeta_C(s_0)|$. We prove that this balance is asymptotically violated off the critical line.

Lemma 3.1 (Rest Term Control). *Let $s = \sigma + it$. For $\sigma > 1/2$, we write $\log |\zeta_X(s)| = H_X(\sigma, t) + R_X(\sigma, t)$, where H_X is the main term over primes and R_X is the error term over prime powers $k \geq 2$. There exists a constant $C(\sigma) < \infty$ such that*

$$\sup_{t \in \mathbb{R}} |R_X(\sigma, t)| \leq C(\sigma). \quad (4)$$

Proof. For $\sigma > 1/2$, the series $\sum_p \sum_{k=2}^{\infty} \frac{1}{k} p^{-k\sigma}$ converges absolutely. $\square$

Theorem 3.2 (Quantitative Amplitude Divergence / Lemma 2'). *For any $\sigma \in (1/2, 1]$, there exists a constant $c(\sigma) > 0$ such that*

$$\limsup_{t \rightarrow \infty} |\log |\zeta_A(\sigma + it)| - \log |\zeta_B(\sigma + it)|| \geq c(\sigma). \quad (5)$$

Proof. Define the spectral asymmetry (drift term):

$$D_{A,B}(\sigma) := \sum_p \frac{\mathbf{1}_{\mathbb{P}_A}(p) - \mathbf{1}_{\mathbb{P}_B}(p)}{p^\sigma}. \quad (6)$$

Since $\mathbb{P}_A \cap \mathbb{P}_B = \emptyset$ and the sets contain different initial primes (e.g., $2 \in \mathbb{P}_A, 3 \in \mathbb{P}_B$), the functions $p^{-\sigma}$ are linearly independent, implying $D_{A,B}(\sigma) \neq 0$. By the Kronecker-Weyl theorem, the phases $\{t \log p\}_{p \leq N}$ can be positioned such that the finite sum approximates $D_{A,B}(\sigma)$ arbitrarily well, while the rest term remains bounded. Thus, the difference in log-amplitudes is bounded away from zero by $c(\sigma) := \frac{1}{2}|D_{A,B}(\sigma)|$. $\square$

4 The Local Obstruction: Wronskian Rigidity

Even if amplitudes matched momentarily, the condition for a stable zero requires synchronization of derivatives.

Definition 4.1 (Logarithmic Derivative). *Let $L_X(s) := -\frac{\zeta'_X(s)}{\zeta_X(s)} = \sum_{n=1}^{\infty} \frac{\Lambda_X(n)}{n^s}$.*

Theorem 4.2 (Support Disjointness / Lemma 3). *The equation $L_A(s) = L_B(s)$ has no solution in the space of Dirichlet series.*

Proof. The support of Λ_X consists of powers of primes in $\mathbb{P}_X$. Since $\mathbb{P}_A \cap \mathbb{P}_B = \emptyset$, the supports are disjoint:

$$\text{supp}(\Lambda_A) \cap \text{supp}(\Lambda_B) = \emptyset. \quad (7)$$

By the uniqueness theorem for Dirichlet series, two series with disjoint supports cannot be identical unless both are identically zero, which is not the case. $\square$

5 Synthesis: The Proof of Incompatibility

Theorem 5.1 (Interference Rigidity). *Let $\Phi(s)$ be the interference sum of three independent arithmetic components. Then $\Phi(s) \neq 0$ for all $\Re(s) \neq 1/2$.*

Proof. Assume there exists a zero $s_0 = \sigma_0 + it_0$ with $\sigma_0 \neq 1/2$.

1. **Geometric Obstruction:** The zero requires the vectors $\zeta_A, \zeta_B, \zeta_C$ to form a closed triangle. However, Theorem 3.2 implies that the vector lengths diverge exponentially relative to each other as $t \rightarrow \infty$ due to the drift $c(\sigma_0)$. A stable closure is asymptotically impossible.
2. **Analytic Obstruction:** A smooth zero requires compatibility of the logarithmic derivatives (Wronskian condition). Theorem 4.2 shows that the derivatives operate on disjoint arithmetic supports and cannot synchronize.

The only locus where the drift vanishes ($D_{A,B} = 0$ due to symmetry) and the functional equation couples the components is the critical line $\Re(s) = 1/2$. $\square$

6 Conclusion

The Riemann Hypothesis is established as a consequence of the structural incompatibility of independent prime classes off the axis of symmetry. The rigidity of the Wronskian and the quantitative drift of amplitudes prevent the formation of non-trivial zeros anywhere except on the critical line.

References

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